Student Table

PART 2 — INTERMEDIATE

Q1. Enable versioning on the personal column family — keep up to 5 versions.

```
hbase(main):021:0> alter 'student', NAME => 'personal', VERSIONS => 5
Updating all regions with the new schema...
1/1 regions updated.
Done.
Took 2.7360 seconds
hbase(main):022:0>
```

Q2. Update personal:city for student 2 (sara) three times: first to cairo, then giza, then back to alex. Then retrieve all versions of that cell.

```
Done.
Took 2.7360 seconds
hbase(main):022:0> put 'student', '2', 'personal:city', 'cairo'
Took 0.0099 seconds
hbase(main):023:0> put 'student', '2', 'personal:city', 'giza'
Took 0.0084 seconds
hbase(main):024:0> put 'student', '2', 'personal:city', 'alex'
Took 0.0077 seconds
hbase(main):025:0>
hbase(main):026:0* get 'student', '2', {COLUMN => 'personal:city', VERSIONS => 3}
COLUMN                              CELL
 personal:city                      timestamp=2026-05-24T11:02:23.007, value=alex
 personal:city                      timestamp=2026-05-24T11:02:22.956, value=giza
 personal:city                      timestamp=2026-05-24T11:02:22.893, value=cairo
1 row(s)
Took 0.0650 seconds
hbase(main):027:0>
```

Q3. Use SingleColumnValueFilter to find all students in year 4.

```
hbase(main):027:0> scan 'student', {
hbase(main):028:1*   FILTER => "SingleColumnValueFilter('academic','year',=,'binary:4')"
hbase(main):029:1> }
ROW                                  COLUMN+CELL
 2                                   column=academic:faculty, timestamp=2026-05-23T13:46:26.169, value=medicine
 2                                   column=academic:grade, timestamp=2026-05-23T13:46:26.441, value=B
 2                                   column=academic:year, timestamp=2026-05-23T13:46:26.639, value=4
 2                                   column=contact:email, timestamp=2026-05-24T10:42:52.222, value=sara@uni.edu
 2                                   column=personal:age, timestamp=2026-05-23T13:45:56.878, value=22
 2                                   column=personal:city, timestamp=2026-05-24T11:02:23.007, value=alex
 2                                   column=personal:name, timestamp=2026-05-23T13:45:56.140, value=sara
 4                                   column=academic:faculty, timestamp=2026-05-23T13:46:26.317, value=arts
 4                                   column=academic:grade, timestamp=2026-05-23T13:46:26.514, value=C
 4                                   column=academic:year, timestamp=2026-05-23T13:46:26.707, value=4
 4                                   column=personal:age, timestamp=2026-05-23T13:45:57.063, value=23
 4                                   column=personal:city, timestamp=2026-05-23T14:02:22.840, value=mansoura
 4                                   column=personal:name, timestamp=2026-05-23T13:45:56.285, value=nada
2 row(s)
Took 0.0719 seconds
hbase(main):030:0>
```

Q4. Use SingleColumnValueFilter to find all students who live in cairo.

```
hbase(main):030:0> scan 'student', {
hbase(main):031:1*   FILTER => "SingleColumnValueFilter('personal','city',=,'binary:cairo')"
hbase(main):032:1> }
ROW                                  COLUMN+CELL
 1                                   column=academic:grade, timestamp=2026-05-24T10:25:35.860, value=A+
 1                                   column=academic:year, timestamp=2026-05-23T13:46:26.607, value=3
 1                                   column=contact:email, timestamp=2026-05-24T10:42:52.142, value=ali@uni.edu
 1                                   column=personal:age, timestamp=2026-05-23T13:45:56.828, value=21
 1                                   column=personal:city, timestamp=2026-05-23T13:45:56.458, value=cairo
1 row(s)
Took 0.0206 seconds
hbase(main):033:0>
hbase(main):034:0*
```

Q5. Add contact:phone for students 1, 3, and 4 with any phone numbers you choose.

```
hbase(main):033:0>
hbase(main):034:0* put 'student', '1', 'contact:phone', '01011111111'
Took 0.0170 seconds
hbase(main):035:0> put 'student', '3', 'contact:phone', '01022222222'
Took 0.0063 seconds
hbase(main):036:0> put 'student', '4', 'contact:phone', '01033333333'
Took 0.0175 seconds
hbase(main):037:0>
hbase(main):038:0*
```

Q6. Scan the table and return only the contact column family for all students

```
hbase(main):038:0*  scan 'student', {COLUMN => 'contact'}
ROW                                     COLUMN+CELL
 1                                      column=contact:email, timestamp=2026-05-24T10:42:52.142, value=ali@uni.edu
 1                                      column=contact:phone, timestamp=2026-05-24T11:04:51.854, value=01011111111
 2                                      column=contact:email, timestamp=2026-05-24T10:42:52.222, value=sara@uni.edu
 3                                      column=contact:phone, timestamp=2026-05-24T11:04:51.954, value=01022222222
 4                                      column=contact:phone, timestamp=2026-05-24T11:04:52.011, value=01033333333
4 row(s)
Took 0.0359 seconds
hbase(main):039:0>
```

Q7. Add a new column family called enrollment to the student table, then verify it appears in the schema.

```
hbase(main):039:0> alter 'student', NAME => 'enrollment'
Updating all regions with the new schema...
1/1 regions updated.
Done.
Took 2.1359 seconds
hbase(main):040:0>
hbase(main):041:0* describe 'student'
Table student is ENABLED
student
COLUMN FAMILIES DESCRIPTION
{NAME => 'academic', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '3', KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE',
 COMPRESSION => 'NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE => '65536', REPLICATION_SCOPE => '0'}

{NAME => 'contact', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1', KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE',
COMPRESSION => 'NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE => '65536', REPLICATION_SCOPE => '0'}

{NAME => 'enrollment', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1', KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE
', COMPRESSION => 'NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE => '65536', REPLICATION_SCOPE => '0'}

{NAME => 'personal', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '5', KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE',
 COMPRESSION => 'NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE => '65536', REPLICATION_SCOPE => '0'}

4 row(s)

QUOTAS
0 row(s)
Took 0.0776 seconds
hbase(main):042:0>
```

Q8. CHALLENGE ⍰ Write a scan that returns only students who are in the engineering faculty AND in year 3.

```
Took 0.0129 seconds
hbase(main):050:0> scan 'student', {
hbase(main):051:1*  FILTER => "SingleColumnValueFilter('academic','faculty',=,'binary:engineering') AND SingleColumnValueFilter('academic','yea
r',=,'binary:3')"
hbase(main):052:1> }
ROW                                    COLUMN+CELL
 1                                     column=academic:grade, timestamp=2026-05-24T10:25:35.860, value=A+
 1                                     column=academic:year, timestamp=2026-05-23T13:46:26.607, value=3
 1                                     column=contact:email, timestamp=2026-05-24T10:42:52.142, value=ali@uni.edu
 1                                     column=contact:phone, timestamp=2026-05-24T11:04:51.854, value=01011111111
 1                                     column=personal:age, timestamp=2026-05-23T13:45:56.828, value=21
 1                                     column=personal:city, timestamp=2026-05-23T13:45:56.458, value=cairo
1 row(s)
Took 0.0542 seconds
hbase(main):053:0>
```